

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** Mechanical Engineering
2. **Subject Name:** Kinematics and Kinetics of Machines
L-T-P: 3-1-0 **Credits:** 4
3. **Pre-requisites:** Basic Engineering Mechanics (ME11003), Dynamics (ME21205)
4. **Syllabus and reference books:**

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Commented [U2]: Please enter the details (use Arial 12 font)

Commented [U3]: Please enter the pre-requisites (subject name and subject no. in Arial 12 font)

Commented [U4]: 1. Please type the syllabus and reference books in the box provided below.

Syllabus:

Introduction to mechanisms: kinematic pairs and their classification, kinematic diagrams, degree of freedom calculation, classification of kinematic chains, kinematic inversions, equivalent linkages, number synthesis, displacement analysis and Grashof criterion, velocity analysis and Kennedy-Aronhold theorem, acceleration analysis;

Dimensional synthesis of planar mechanisms: two and three position synthesis for function generation, path generation and rigid body guidance; dead center problems, branch and order defects;

Cams: nomenclature, displacement diagrams, synthesis of translating flat-face, roller follower cams;

Gears: fundamental law of gearing, nomenclature, characteristics of involutive action, analysis of gear trains, manual and automatic gear boxes;

Introduction to robot kinematics: rotation matrix, Denavit-Hartenberg parameters, homogeneous transformation matrix, robot kinematics;

Vibration analysis: Free and forced vibrations of undamped and damped multi-degree of freedom systems, modes of vibrations, estimation of natural frequencies and damping, base-excited vibrations, vibration transmission and isolation, concept of vibration absorber;

Dynamics of rotating machinery: Analysis of rotor with unbalance, gyroscopic effect, critical speeds, dynamics of a single cylinder engine, turning moment diagram, flywheel design, balancing of rotors and engines

Please use "Arial" font with a font size of "12" and a line spacing of 1.0 for syllabus and reference books. Please use Justified Text (Ctrl+J) for the content entered in the boxes
Reference books should be numbered as 1), 2), 3)
Please provide all bibliographic details for the reference books

Reference Books:

- 1) Theory of Mechanisms and Machines: A. Ghosh and A.K. Mallik
- 2) Kinematics, Dynamics and Design of Machinery: K.J. Waldron and G.L. Kinzel
- 3) Fundamentals of Vibrations: L. Meirovitch

5. **Lecture-wise break-up:**

Commented [U5]: Please provide lecture-wise break-up in the table given below:

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Commented [U6]:

Sl. No.	Topic	No. of lectures
1.	Introduction to mechanisms: kinematic pairs and their classification, kinematic diagrams, degree of freedom calculation, classification of kinematic chains, kinematic inversions, equivalent linkages, number synthesis, displacement analysis and Grashof criterion, velocity analysis and Kennedy-Aronhold theorem, acceleration analysis	5
2.	Dimensional synthesis of planar mechanisms: two and three position synthesis for function generation, path generation and rigid body guidance; dead center problems, branch and order defects	5
3.	Cams: nomenclature, displacement diagrams, synthesis of translating flat-face, roller follower cams	4
4.	Gears: fundamental law of gearing, nomenclature, characteristics of involutive action, analysis of gear trains, manual and automatic gear boxes	5
5.	Introduction to robot kinematics: rotation matrix, Denavit-Hartenberg parameters, homogeneous transformation matrix, robot kinematics	5
6.	Vibration analysis: Free and forced vibrations of undamped and damped multi-degree of freedom systems, modes of vibrations, estimation of natural frequencies and damping, base-excited vibrations, vibration transmission and isolation, concept of vibration absorber	8
7.	Dynamics of rotating machinery: Analysis of rotor with unbalance, gyroscopic effect, critical speeds, dynamics of a single cylinder engine, turning moment diagram, flywheel design, balancing of rotors and engines	8
Total number of hours		40

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